

Yamaha
R1 – R6
2004 – 2011



INTRODUCTION

AIM EVO4, ECU Bridge and SoloDL loggers can communicate with Yamaha R1-R6 2004-2011 ECU using the K line and connecting GND too.

In this document are shown Yamaha and loggers pinout.

Please note: for Yamaha bikes to output data through the K-line the stock dash is to be left on the bike.

Warning: once AIM logger installed it needs to be configured through Race Studio 2 software: refer to each bike paragraph to know the configuration to be set.

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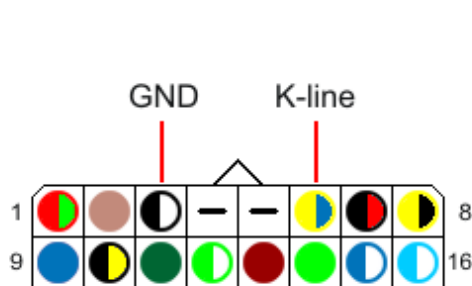
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Chapter 1 – Yamaha K-line

Yamaha R1-R6 2004-2011 bikes are equipped with different stock dash connectors. Here follow images showing where K-line is placed on the different connectors.

1.1 – Yamaha R1 2004-2008

The image here below shows the bikes stock dash connector and its pinout.

							
Pin	Function	Pin	Function				
1	+Vb	9	+Vb				
2	+Vb Ext	10	High-beam				
3	GND	11	Direction light				
4	Not used	12	Fuel alarm				
5	Not used	13	Direction Light				
6	K-line	14	P1				
7	Oil level	15	P2				
8	RPM	16	Neutral				

Race Studio configuration to be set is:

- ECU Manufacturer: Yamaha
- ECU Model: BIKE_R1_04_05_06 for R1 2004-2005-2006 or
- ECU Model: BIKE_R1_07_08 for R1 2007-2008

1.2 – Yamaha R1 2009-2011

The image here below shows the bikes stock dash connector and its pinout.

Images are shown on the left and the labels on the right. See the color code and the pin number.

The diagram shows a 20-pin connector with pins numbered 1 to 20. The pins are color-coded as follows:

- Pin 1: Light blue
- Pin 2: Red
- Pin 3: Light blue
- Pin 4: Black with a red diagonal line
- Pin 5: Black with a yellow diagonal line
- Pin 6: Grey
- Pin 7: Orange
- Pin 8: Red with a black diagonal line
- Pin 9: Yellow
- Pin 10: Black with a yellow diagonal line
- Pin 11: Black with a yellow diagonal line
- Pin 12: Green
- Pin 13: Red with a green diagonal line
- Pin 14: Brown
- Pin 15: Green
- Pin 16: Black with a white diagonal line
- Pin 17: Black with a white diagonal line
- Pin 18: Yellow with a black diagonal line
- Pin 19: White
- Pin 20: Pink

Red lines point from the labels "GND" and "K-line" to pins 16 and 18, respectively.

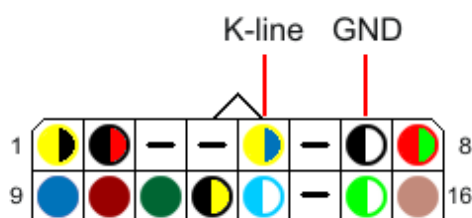
Pin	Function	Pin	Function
1	Not used	11	High beam
2	Dir. light left	12	Dir. light right
3	Not Used	13	Vb Dir
4	Not connected	14	Vb key
5	Switch oil Level	15	Fuel alarm
6	RPM	16	GND
7	6 th gear	17	Led immobilizer
8	5 th gear	18	K-line
9	4 th gear	19	1 st gear
10	3 rd gear	20	2 nd gear

Race Studio configuration to be set is:

- ECU Manufacturer: Yamaha;
- ECU Model: BIKE_R1_09_11_06 for R1 2009-2010-2011.

1.3 – Yamaha R6 2004-2005

The image here below shows the bikes stock dash connector and its pinout.



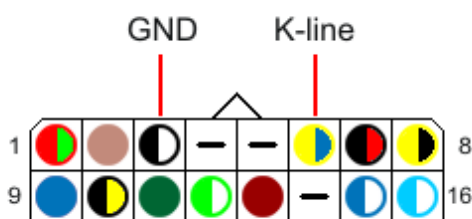
Pin	Function	Pin	Function
1	RPM	9	+Vb
2	Oil Level	10	Direction light
3	Not Used	11	Direction light
4	Not used	12	High-beam
5	K-line	13	Neutral
6	Not used	14	Not used
7	GND	15	Fuel
8	+Vb	16	+Vb Ext

Race Studio configuration to be set is:

- ECU Manufacturer: Yamaha;
- ECU Model: BIKE_R6_04_05 for R6 2004-2005.

1.4 – Yamaha R6 2006-2011

The image here below shows the bikes stock dash connector and its pinout.



Pin	Function	Pin	Function
1	+Vb	9	+Vb
2	+Vb Ext	10	High-beam
3	GND	11	Direction light
4	Not used	12	Fuel alarm
5	Not used	13	Direction light
6	K-line	14	P1
7	Oil level	15	P2
8	RPM	16	Neutral

Race Studio configuration to be set is:

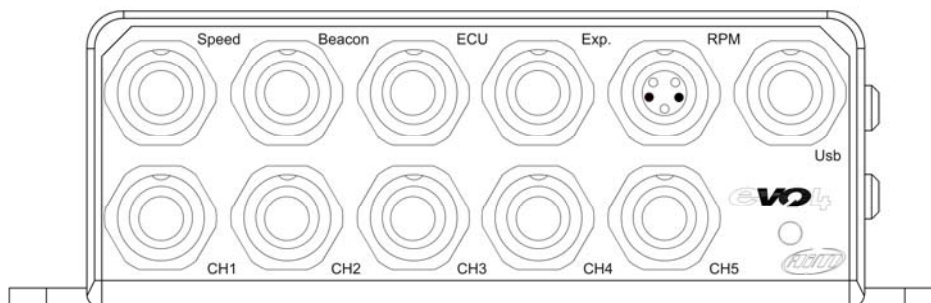
- ECU Manufacturer: Yamaha;
- ECU Model: BIKE_R6_06_10 for R6 2006-2007-2008-2009-2010-2011.

Chapter 2 – AIM loggers K-line

AIM EVO4, ECU Bridge and Solo DL are all equipped with the K-line. Here follow images showing their pinout.

2.1 – EVO4

K-Line is on pin 4 of EVO4 RPM connector while GND is on pin 2 of the same connector as shown here below.



2.2 – SoloDL

SoloDL is equipped with two bottom connectors. K-line is on pin 6 of the 7 pins female connector while GND is on pin 7.



2.3 – ECU Bridge

AIM ECU Bridge is available in two versions but only the one with OBDII connector is equipped with the K-line. The K line is on pin 7 of OBDII connector while GND is on pin 5. Please note: for information about how to connect ECU Bridge to Yamaha ECU contact AIM support at support@aim-sportline.com.

Chapter 3 – Communication Protocol

Channels received by EVO4, ECU Bridge and SoloDL connected to Yamaha ECU are:

ID	CHANNEL NAME	FUNCTION
ECU_1	R1_RPM	RPM
ECU_2	R1_WH_SPEED	Wheel speed
ECU_3	R1_ECU_ERR	ECU error
ECU_4	R1_WATER_TEMP	Water Temperature
ECU_5	R1_AIR_TEMP	Air Temperature